Differential Effects of Dietary Ingestion of White and Brown Beans (*Phaseolus Vulgaris L*) on Lipid Profile and Adipose Tissue Distribution in Male Wistar Rats

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ABSTRACT

This study investigated the impact of dietary consumption of white and brown beans (*Phaseolus vulgaris L*) on lipid profile and adipose tissue distribution in rats. Twenty-five male Wistar rats of comparable weights (100 ± 20g) were assigned to five groups; a control, fed normal rat chow, test groups 2 and 3, fed 25% and 50% white beans- and 4 and 5, fed 25% and 50% brown beans- diets for 21 days. By 22nd day of diet consumption, blood samples were collected in plain bottles and centrifuge at 3000 rpm for 10 minutes to obtain serum for total cholesterol, triglycerides, and high-density lipoprotein estimation using commercial kits while low density lipoprotein (LDL) was calculated based using standard formula. The rats were dissected to obtain and estimated thoracic and abdominal adipose tissues weight using electronic weighing scale. The result showed a significant decreased in body weight gain in the beans-fed groups. There was improved lipid profile, with significant decreased in blood glucose (50% white beans), serum triglycerides (50% white beans), LDL cholesterol (50% brown beans), and abdominal adipose tissue weight (50% white and brown bean) compared with the control. This study suggests that consumption of white or brown beans, especially at moderate to high dietary levels, may support metabolic health by modulating lipid metabolism, glucose homeostasis, and adipose tissue accumulation.

Keywords: *Phaseolus vulgaris*, Adipose tissue, Low density lipoprotein, Triglyceride, cholesterol,

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1. INTRODUCTION

The global rise in metabolic disorders and non-communicable chronic diseases such as obesity, dyslipidemia, and type 2 diabetes has necessitated increased research into dietary factors that influence lipid metabolism, glucose homeostasis, and adipose tissue distribution [1-4]. These conditions are strongly linked to unhealthy dietary habits, physical inactivity, and genetic predispositions, and are major contributors to morbidity and mortality worldwide [1, 3, 4]. Modifying dietary patterns remains one of the most practical and cost-effective strategies to prevent and manage these disorders [1, 4].

Legumes, particularly beans (*Phaseolus vulgaris* L), have received notable attention in nutritional science due to their health-promoting properties. Its local names in Nigeria are “ewa” (Yoruba), “wanke” (Hausa) and “egwa” (Igbo). Beans are staple foods in many parts of the world and are recognized for their high content of plant-based proteins, complex carbohydrates, dietary fiber, vitamins, and minerals [5]. In addition, they contain a variety of bioactive compounds such as phenolics and flavonoids that have been shown to exert antioxidant, hypolipidemic, and glycemic-modulating effects [6, 7].

Among the different varieties of beans, white and brown beans are commonly consumed in both traditional and modern diets. Although they belong to the same species, their nutritional compositions differ, particularly in pigment-associated phytochemicals. These

common beans are generally richer in polyphenols and flavonoids, and do not differe mostly in their nutritiobal compositions, they however, differ slightly in taste, texture and cooking time [8]. Brown been have been shown to modulate lipid absorption, oxidative stress, fat metabolism and possess anti-fungal, anti-diabtic and anti-tumor activities [9, 10]. On the other hand, white beans have a different profile of resistant starch and fiber, which have been documented to influence the body's metabolism, regulate serum lipid profile, inhibit precancerous growth [11, 12] and reduce inflammatory biomarkers [13]. Despite these differences, there is limited comparative research examining how these bean types differentially affect lipid profiles, blood glucose regulation, and fat distribution.

Fat accumulation in specific body regions, particularly in the abdominal (visceral) and thoracic (subcutaneous) compartments, plays an important role in determining metabolic health [14]. Visceral adiposity is closely linked to insulin resistance, systemic inflammation, and increased cardiovascular risk, whereas subcutaneous fat, particularly in the thoracic region, may have less detrimental or even protective metabolic effects [14]. Thus, evaluating not just overall body weight but also regional fat deposition is essential in assessing the metabolic impacts of different diets.

This study, therefore, seeks to investigate the differential effects of dietary consumption of white and brown beans on body weight, lipid profile, blood glucose, and adipose tissue distribution in male Wistar rats. By focusing on both the mean thoracic and abdominal adipose tissue weights, this study aims to elucidate how these two bean varieties influence not only total fat accumulation but also its regional distribution, a factor increasingly recognized for its clinical significance. The results of this research may provide insights into the metabolic benefits or drawbacks of white versus brown beans consumption. Such findings can inform dietary recommendations, contribute to the development of functional food products, and guide future studies exploring plant-based interventions for metabolic disorders.

2. MATERIALS AND METHODS

Material: Two different species of beans; white beans also referred to as Sokoto beans, and brown beans also known as honey beans, were purchased from Ekpoma local market, Edo State, Nigeria. Rat chow (Bendel feeds and flour meal Ewu, Nigeria) was obtained from a local store in Ekpoma. All other chemicals were of analytical grade.

Care and management of Wistar rats: Twenty-five male Wistar rats of comparable weights (100 ± 20g) were used for this study. The rats were obtained from the Animal House of the College of Medicine, Ambrose Alli University, Ekpoma and transferred to the Department of Physiology animal holding facility in the same institution. They were housed in plastic cage with a wire screen top. The animals were fed *ad libitum* with clean tap water and pelleted rat chow during acclimatisation (7 days) and were maintained under standard care of laboratory animals as stated in the guidelines and regulations for the use of animals in scientific experiment.

Diet formulation: A bowl each of white and brown beans were wash with clean water separately. Thereafter, they were put in iron pot and heated for 50 to 60 minutes adding adequate amount of water (3.5 to 5.5 litres) to cook the beans. After cooking, they were allowed to cool and then air dried. After drying they were grinded using electronic machine into fine coarse powder and were used for the diets formulations.

The preparation of the 25% and 50% of the white and brown bean diets was as described by Nwaopara *et al.*, [15]. Briefly, to prepare 3000g of the 25% white or brown beans diet, 750g of the grinded white or brown beans was mixed with 2250g of normal rat chow and for the 50% white or brown beans diet, 1500g of the grinded white or brown beans was mixed

with equal quantity of normal rat chow. These were thoroughly mixed by adding adequate clean water to form paste. The pastes were cut into small pieces (approximately 2-4 mm diameter) and allowed to dry under the sun. The sun-dried mixture was then packed into clean sack bag for use.

Experimental grouping: After acclimatization, the rats were assigned to five (5) groups.

Group 1 (control group): fed normal rat chow (100g daily) and clean water.

Group 2 (25% white beans diet group): fed formulated 25% white beans diet (100g daily) and clean water.

Group 3 (50% white beans diet group): fed formulated 50% white beans diet (100g daily) and clean water.

Group 4 (25% brown beans diet group): fed formulated 25% brown beans diet (100g daily) and clean water.

Group 5 (50% brown beans diet group): fed formulated 50% brown beans diet (100g daily) and clean water.

Duration of feeding: Rats in all the groups received *ad libitum* the formulated diets. They were giving the different diets daily between the hours of 8:00 am and 10:00 am. Next day left over diets and water were discarded while rats bedding was changed every other day. The administration of the different beans' diets lasted for 21 days.

Sample collection: At the 22nd day of diet administration, the rats were cervically decapitated and blood sample collected following standard laboratory procedures. Briefly, the rats were dissected along the lateral side of the neck to assess the jugular blood vessels. Blood sample was then obtained using 2mL syringe and the blood collected into plain bottles.

Sample analysis: The blood samples were centrifuge at 3000 rpm for 10 minutes to obtain the serum. Serum concentration of total cholesterol (TC), triglycerides (TG), and high-density lipoprotein (HDL) concentrations were determined using commercial kits while low density lipoprotein (LDL) was calculated based on standard formula. The rats were dissected to obtained thoracic and abdominal adipose tissues and were weighed using electronic weighing balance. The concentration of LDL was calculated from the Fiedewald's formula: LDL = TC – HDL – TG /5.

Data analysis: The mean ± standard error of mean was determined and the one-way ANOVA statistical test was performed using SPSS version 20 software. The value $p < 0.05$ was considered statistically significant. Results were then presented in suitable charts.

3. RESULTS

1. Changes is body weight in rats fed white and brown species of beans

Figure 1 is a comparative chart of the changes is body weight in adult Wistar rats fed white and brown species of beans. White or brown beans ingestions resulted in a decrease in percentage body weight gain over the 21 days of the study. Statistically, the 25% and 50% white or brown beans resulted in significant decrease in percentage body weight compared with the control diet.

2. Serum lipid profile in rats fed white and brown species of beans.

Figure 2 is a comparative chart of lipid profile levels in rats fed white and brown species of beans. There was no significant change ($p>0.05$) in serum total cholesterol level with the ingestion of white or brown beans compared with the control diet. White bean resulted in decrease in serum triglyceride level with significant decrease ($p<0.05$) in the 50% white beans group compared with the control. The brown bean did not significantly alter

(p>0.05) serum triglyceride level compared with the control diet. There was no significant change in serum HDL level following the ingestion of white or brown beans compared with the control. The ingestion of white or brown bean resulted in a decrease in LDL cholesterol level with significant decrease (p<0.05) in the 50% brown beans compared with the control diet.

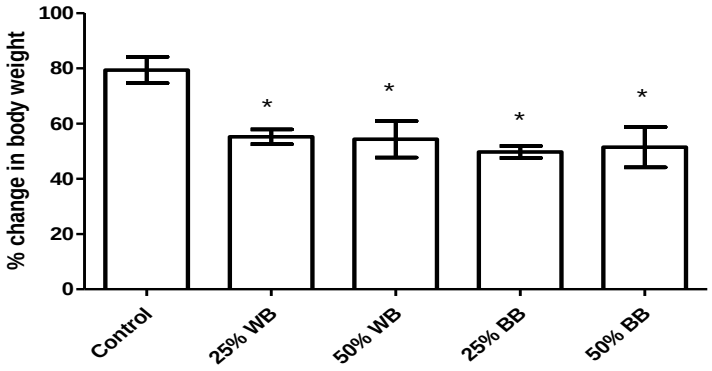


Figure 1: Changes is body weight in rats fed white and brown species beans.

Values are means ± SEM, n = 5, *significant compared with control. Key: 25% WB = 25% of white beans, 50% WB = 50 % of white beans, 25% BB = 25% of brown beans, 50% BB = 50% of brown beans.

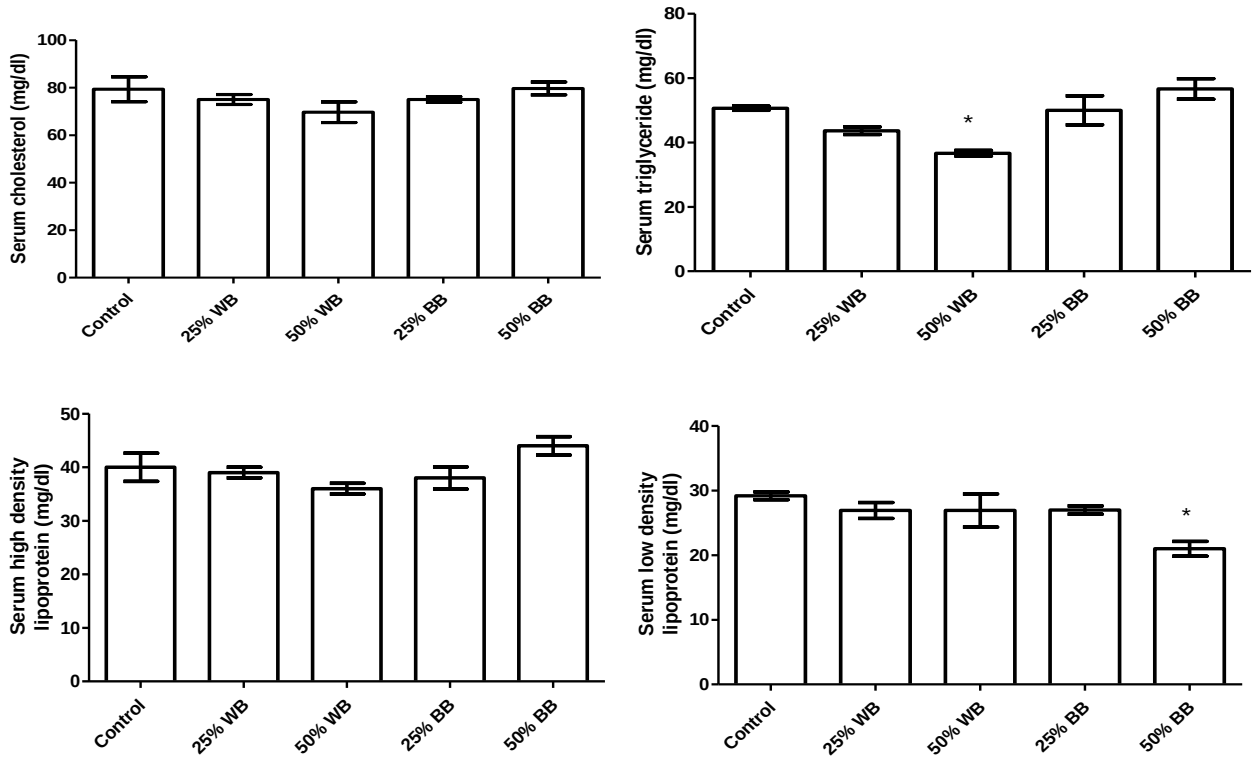


Figure 2: Serum lipid profile in rats fed white and brown species of beans.

Values are means ± SEM, n = 5, *statistically significant compared with control diet. Key: 25% WB = 25% of white beans, 50% WB = 50 % of white beans, 25% BB = 25% of brown beans, 50% BB = 50% of brown beans.

3. Blood glucose and thoracic and abdominal adipose tissue weight in rats fed white and brown species of beans.

Figure 3 represents the blood glucose and adipose tissue weight in rats fed white and brown species of beans. White or brown beans ingestion cause a decrease in blood glucose over the period of study with significant decrease in the 50% white beans fed group compared with the control. Mean thoracic adipose tissue weight was non-significantly decreased following ingestion of white or brown bean. Mean abdominal adipose tissue weight was observed to decrease following the ingestion of white or brown beans with significantly decrease ($p<0.05$) in 50% white and brown beans fed groups compared with the control.

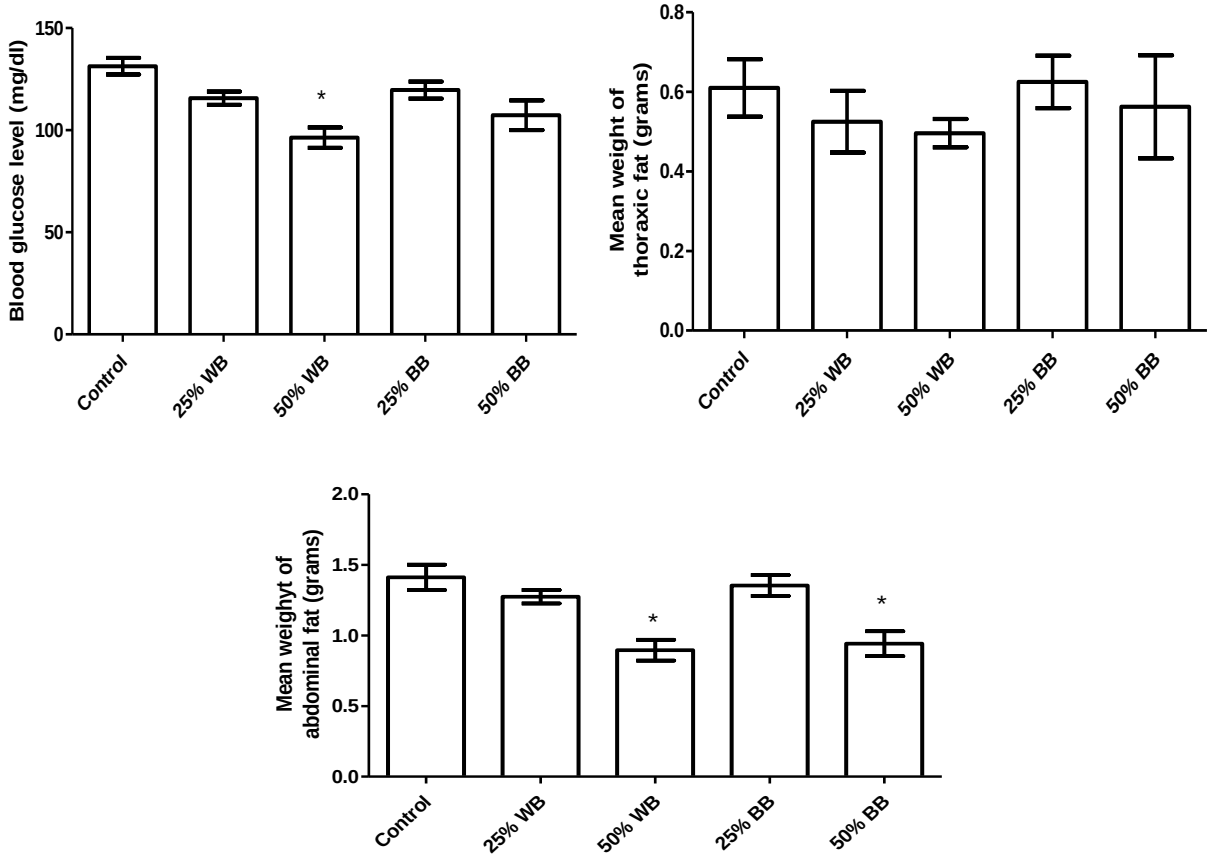


Figure 3: Evaluation of (a) blood glucose, (b) mean thoracic adipose tissue weight and (c) mean abdominal adipose tissue weight in rats fed white and brown species of beans.

Values are means ± SEM, n = 5, *significant compared with control diet. Key: 25% WB = 25% of white beans, 50% WB = 50 % of white beans, 25% BB = 25% of brown beans, 50% BB = 50% of brown beans.

4. DISCUSSION

This study examined the comparative impact of dietary consumption of white and brown beans on lipid profile, body weight, blood glucose, and adipose tissue distribution in adult male Wistar rats. The findings provide valuable insights into the potential metabolic benefits associated with consuming different varieties of beans. The consumption of both white and brown beans resulted in a reduction in percentage body weight gain, serum

triglyceride, low density lipoprotein, blood glucose, and abdominal fat. This outcome may be attributed to the high fiber content and low glycemic index of *Phaseolus vulgaris*, which has been associated with improved satiety and reduced caloric intake as documented by Bazzano *et al*., [16]. Additionally, bioactive compounds in beans, such as phenolics, flavonoids, and resistant starch, may contribute to reduced energy absorption and fat deposition [17].

The reducing effect on serum triglyceride and LDL, suggest a triglyceride-lowering effect potentially mediated through modulation of hepatic lipogenesis or enhanced lipid clearance. Tannins and polyphenols known to be a composition of beans have been reported to influence cholesterol metabolism or uptake in the liver [18]. Although HDL levels were unaffected, the reduction in LDL and triglycerides, especially in higher inclusion levels, indicates a positive shift in lipid metabolism.

Both seeded beans showed decrease in blood glucose levels. This effect may be linked to the hypoglycemic properties of beans, which are known to slow glucose absorption due to their fiber content, α-amylase inhibitors, and other anti-diabetic constituents [19]. These findings support the role of *Phaseolus vulgaris* as a functional food in glycemic control, especially in the context of metabolic disorders.

Thoracic and abdominal adipose tissue weights were reduced following ingestion of both seeded beans. Since abdominal fat is more metabolically active and closely linked with insulin resistance and cardiovascular risk, the reduction observed may indicate a beneficial shift toward improved metabolic health. The non-significant change in thoracic adipose tissue suggests a site-specific sensitivity of adipose depots to dietary interventions, with abdominal fat being more responsive to bioactive food components..

CONCLUSION

These results suggest that incorporating *Phaseolus vulgaris*, especially at moderate to high dietary levels, may support metabolic health by modulating lipid metabolism, glucose homeostasis, and adipose tissue accumulation. The differences observed between white and brown beans may be due to variations in their phytochemical composition and bioactive profiles, warranting further biochemical and mechanistic studies.

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